👩‍🏫 Staff Guide · Small Engines Overhaul · Unit 8 Engine Teardown & Rebuild (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Weeks 6-9+: engine project (teardown and rebuild), Ring End Gap/Land Clearance Handout, How to grind valves video, Engine Assembly 1, iCADEMY tech bits (engine quick check, rewind starter repair, valve clearance, throttle cable, crankcase sealing, chain, trimmer, blower and cut-off saw service, chaps)")	**Level 2 piece**
Engine project (teardown/rebuild, not written in Canvas)	Teardown & Rebuild lab packet: step-by-step picture checklist
Ring End Gap / Land Clearance Handout	Notes B + Ring Gap steps in the lab packet
How to grind valves video	Video guide (link only)
Engine Assembly 1	Lab packet reflection + checklist (photo proof)
iCADEMY tech bits (engine check, rewind starter, valve clearance, chain/trimmer/blower/cut-off saw, chaps)	Slides 'Outdoor Power Equipment Service' + notes C (our own content)
Class slides	Deck: 17 slides, Words to Know first, hold-up-1-2-3 checks, recap
His worksheets (grouped into lessons)	Guided notes: 11 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no "all of the above"
- Labeled parts tray (muffin tin) and a printed photo checklist at the bench
- Pair with a gen-ed partner; student owns specific steps
- Staff demonstrate each new tool first; student uses it with adult supervision

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part

4. What to grade

- Engine project: grade on the checklist steps completed with photo proof, not speed
- Troubleshooting: L1 points to fuel, spark, compression, air cards
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in his files say "check with the class teacher".

🚧 Under construction (check with the class teacher)

- **Engine project checklist and grading**: The project steps and rubric are done in class and not written in Canvas; our checklist is team-written.
- **Troubleshooting**: A syllabus goal with no class materials; our lessons are team-written.
- **iCADEMY tech bits**: STIHL iCADEMY is login-only; we wrote our own service content from public knowledge.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.

- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

👩🏾‍🏫 Staff Guide · Engine Teardown & Rebuild (Level 2)

Small Engines Overhaul · Unit 8 · goes with the class “Engine project + Engine Assembly 1 + Ring End Gap Handout” (project points). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Engine project + Engine Assembly 1 + Ring End Gap Handout”. 9 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do your hands go while the tool is turning?
- What should we do when we finish?

3. Modifications

- Task card instead of written directions
- Sentence starters on the reflection
- Partner reads directions aloud

4. What to grade (project points)

- Steps done (photo proof)
- Reflection: 2 sentences per box
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

🚧 Under construction (check with the class teacher)

- **This packet**: The class engine project steps and rubric are not written in Canvas; this checklist is team-written. Specs vary by engine model; staff fill them in from the class manual.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
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- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

🔧 Teardown & Rebuild Notes

Small Engines Overhaul · Unit 8 Engine Teardown & Rebuild · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Teardown

1. Number the teardown steps:

_____ Remove outside parts

_____ Remove the head and inside parts

_____ Disconnect plug wire and drain fluids

_____ Remove the flywheel

2. Small parts go in a labeled parts _____.

Word bank: **tray**

B. Rings and valves

3. Ring end _____ is measured in the cylinder with a feeler gauge.

4. _____ makes the valve seal on its seat.

5. Valve _____ is set with a feeler gauge.

Word bank: **clearance** **gap** **Lapping**

C. Rebuild and troubleshoot

6. Head bolts are tightened to _____ spec.

7. An engine needs fuel, spark, _____ and air.

8. Chainsaw users wear _____.

Word bank: **chaps** **compression** **torque**

Level 2 version of the gen ed “Engine project + Ring End Gap Handout.”

📖 Engine Teardown & Rebuild: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Teardown

Taking an engine apart in order, keeping parts organized and recording each step.



Parts tray

A tray with labeled cups so small parts stay in order during teardown.



Ring end gap

The gap between the ends of a piston ring in the cylinder, checked with a feeler gauge.



Valve clearance

The small gap between the valve stem and the tappet/rocker when the valve is closed.



Lapping valves

Rubbing a valve on its seat with lapping compound so it seals tightly.



Torque

The twisting force on a bolt, set to spec with a torque wrench (ft-lb or in-lb).



Troubleshooting

Checking fuel, spark, compression and air in order to find why an engine won't run right.



Rewind starter

The pull-rope starter with a coiled spring that rewinds the rope.

🔧 Engine Teardown & Rebuild

Small Engines Overhaul · Unit 8 · Level 2



Name: _____

Modified from the class “Engine project + Engine Assembly 1 + Ring End Gap Handout”.

🧺 Tools

- ☐ 🧰 **Socket set + wrenches**
- ☐ 🔘 **Flywheel puller** with adult
- ☐ 🔧 **Torque wrench** with adult
- ☐ 🌀 **Valve spring compressor** with adult

🧺 Supplies

- ☐ 🥣 **Parts tray** labeled cups
- ☐ 📄 **Feeler gauges**
- ☐ 🔄 **Lapping compound**
- ☐ 🗒️ **New gasket set**

🔧 Steps: check each box when you finish

- [ ] 1 🔌 Disconnect plug wire; with adult supervision drain fuel and oil.
- [ ] 2 📸 Photo; remove shroud, air filter, carb, muffler (tray cups 1-4).
- [ ] 3 🛞 Remove flywheel with the puller; remove armature.
- [ ] 4 🔩 Remove head; remove valves with the spring compressor.
- [ ] 5 ⭕ Remove piston/rod; check ring end gap and side clearance; record.
- [ ] 6 🔄 Lap valves; set valve clearance to spec.
- [ ] 7 ⏱️ Reassemble: align timing marks; new gaskets; oil parts.
- [ ] 8 🔧 Torque head bolts in pattern to spec.
- [ ] 9 🔍 Test: fuel, spark, compression; adult starts the engine outdoors.

🚧 **Under construction:** The class engine project steps and rubric are not written in Canvas; this checklist is team-written. Specs vary by engine model; staff fill them in from the class manual. *Check with the class teacher.*

🔧 Engine Teardown & Rebuild · Prep Sheet

Small Engines Overhaul · Unit 8 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🔧 **Mechanic**	Does the main engine job with the tools	
🧰 **Tool helper**	Gets tools from the board and returns them to staff	
📋 **Recorder**	Takes photos and fills in the chart	
🧽 **Clean-up captain**	Wipes the bench, sweeps, checks tools are back	

🧰 Tools and supplies: check when you have it

- ☐ 🧰 **Socket set + wrenches**
- ☐ 🔘 **Flywheel puller** with adult
- ☐ 🥣 **Parts tray** labeled cups
- ☐ 📄 **Feeler gauges** ring gap, valves
- ☐ 🔧 **Torque wrench** with adult
- ☐ 🔄 **Valve lapping stick + compound**
- ☐ 🌀 **Valve spring compressor** with adult
- ☐ 📸 **Camera/iPad** every step

⚠️ Safety and manners focus

🥽 **Safety glasses** on the whole time.

👖 **Coveralls** and closed-toe shoes; tie back long hair.

🔧 Use tools **with adult supervision**. Right tool for the job.

🌀 Valve and starter **springs** store energy: eyes protected, tool held with adult supervision.

🏭 Engines are started **outdoors** with an adult (carbon monoxide).

🧼 **Clean up** and wash hands when you finish.

✅ Before you start

☐ Safety glasses ☐ Coveralls ☐ Adult at bench

🔧 Engine Teardown & Rebuild · Reflection

Small Engines Overhaul · Unit 8 · Level 2



Name: _____

📸 My photos / proof

☐ Before photo ☐ During photo ☐ Finished photo

✅ Skills I used today (check all you did)

☐ 🥽 Wore safety glasses and used tools safely

☐ 🔧 Finished the Engine Teardown & Rebuild steps

☐ 🧽 Cleaned up and returned tools

☐ 🤝 Did my job role

What did you do today, in order?

1. _____
2. _____

Ideas: **First…** **Next…** **Last…**

What did you learn about engines?

1. _____
2. _____

Ideas: **The engine needs…** **I saw…** **Next time…**

✅ Engine Teardown & Rebuild Quiz

Small Engines Overhaul · Unit 8 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🔌 First teardown step:

A. Remove the piston

B. Disconnect plug wire, drain fluids

C. Start it

2. 🔘 The flywheel is removed with a…

A. Hammer

B. Pry bar only

C. Flywheel puller

3. 📸 Photos during teardown help…

A. Reassembly

B. Painting

C. Nothing

4. ⭕ The bottom ring is the…

A. Top compression ring

B. Valve ring

C. Oil control ring

5. 📄 Ring end gap is measured with a…

A. Feeler gauge

B. Ruler

C. Caliper only

6. 🔄 Lapping valves…

A. Makes them bigger

B. Makes them seal

C. Cleans the carb

7. 🔧 Head bolts are tightened…

A. As tight as possible

B. To torque spec, in pattern

C. Randomly

8. ⏱️ Timing marks line up the…

A. Crank and cam gears

B. Tank and cap

C. Wheels

9. 🔍 The four needs of an engine:

A. Paint, oil, wheels, rope

B. Water, salt, sugar, air

C. Fuel, spark, compression, air

10. 🪚 Chainsaw PPE includes…

A. Chaps, gloves, eye and ear protection

B. Shorts and sandals

C. Nothing

🎬 How to Grind (Lap) Valves

Small Engines Overhaul · Unit 8 · Level 2 video guide (modified). YouTube video linked in Canvas: https://www.youtube.com/watch?v=taAP7YoVhGo

Name: _____

Before you watch

Why does a valve need to seal tightly? _____

While you watch: circle the best answer

1. What is put on the valve face?

A. Oil only | B. Lapping compound | C. Paint

2. How is the valve turned?

A. With a drill at full speed | B. By hammer | C. Back and forth with a lapping stick

3. What shows a good seal?

A. An even gray band all around | B. Shiny spots | C. Scratches

4. What is done after lapping?

A. Leave the compound | B. Paint it | C. Clean off all compound

After you watch: finish the sentence

5. After watching, finish the sentence:

Leftover lapping compound is a problem because

_____

Goes with the class “How to grind valves (video notes)” (class points pts). Same points when complete.

🔧 Teardown & Rebuild Notes · ANSWER KEY

Small Engines Overhaul · Unit 8 Engine Teardown & Rebuild · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Teardown

1. Number the teardown steps:

2 Remove outside parts

4 Remove the head and inside parts

1 Disconnect plug wire and drain fluids

3 Remove the flywheel

2. Small parts go in a labeled parts **tray** .

Word bank: **tray**

B. Rings and valves

3. Ring end **gap** is measured in the cylinder with a feeler gauge.

4. **Lapping** makes the valve seal on its seat.

5. Valve **clearance** is set with a feeler gauge.

Word bank: **clearance** **gap** **Lapping**

C. Rebuild and troubleshoot

6. Head bolts are tightened to **torque** spec.

7. An engine needs fuel, spark, **compression** and air.

8. Chainsaw users wear **chaps** .

Word bank: **chaps** **compression** **torque**

Level 2 version of the gen ed “Engine project + Ring End Gap Handout.”
🔑 Key written by our team from the class teacher's materials, please check.

✅ Engine Teardown & Rebuild Quiz · ANSWER KEY

Small Engines Overhaul · Unit 8 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🔌 First teardown step: Engine project

A. Remove the piston	**B. Disconnect plug wire, drain fluids**	C. Start it

2. 🔘 The flywheel is removed with a… Engine project

A. Hammer	B. Pry bar only	**C. Flywheel puller**

3. 📸 Photos during teardown help… Engine project

A. Reassembly	B. Painting	C. Nothing

4. ⭕ The bottom ring is the… Ring handout

A. Top compression ring	B. Valve ring	**C. Oil control ring**

5. 📄 Ring end gap is measured with a… Ring handout

A. Feeler gauge	B. Ruler	C. Caliper only

6. 🔄 Lapping valves… Valve video

A. Makes them bigger	**B. Makes them seal**	C. Cleans the carb

7. 🔧 Head bolts are tightened… Engine Assembly

A. As tight as possible	**B. To torque spec, in pattern**	C. Randomly

8. ⏱️ Timing marks line up the… Engine Assembly

A. Crank and cam gears	B. Tank and cap	C. Wheels

9. 🔍 The four needs of an engine: Troubleshooting

A. Paint, oil, wheels, rope	B. Water, salt, sugar, air	**C. Fuel, spark, compression, air**

10. 🪚 Chainsaw PPE includes… Service topics

A. Chaps, gloves, eye and ear protection	B. Shorts and sandals	C. Nothing

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed “Engine Assembly 1 + Ring End Gap Handout (team-written)”. Score: __ / 10. 🔑
Key written by our team from Mr. Davis's materials, please check.

🎬 How to Grind (Lap) Valves · ANSWER KEY

Small Engines Overhaul · Unit 8 · Level 2 video guide (modified). YouTube video linked in Canvas: https://www.youtube.com/watch?v=taAP7YoVhGo

Name: _____

Before you watch

Why does a valve need to seal tightly? _____

While you watch: circle the best answer

1. What is put on the valve face?

A. Oil only | **B. Lapping compound** | C. Paint

2. How is the valve turned?

A. With a drill at full speed | B. By hammer | **C. Back and forth with a lapping stick**

3. What shows a good seal?

A. An even gray band all around | B. Shiny spots | C. Scratches

4. What is done after lapping?

A. Leave the compound | B. Paint it | **C. Clean off all compound**

After you watch: finish the sentence

5. After watching, finish the sentence:

Leftover lapping compound is a problem because **it is abrasive and wears the engine**

Goes with the class “How to grind valves (video notes)” (class points pts). Same points when complete.
🔑 Key written by our team from the class teacher's materials, please check.